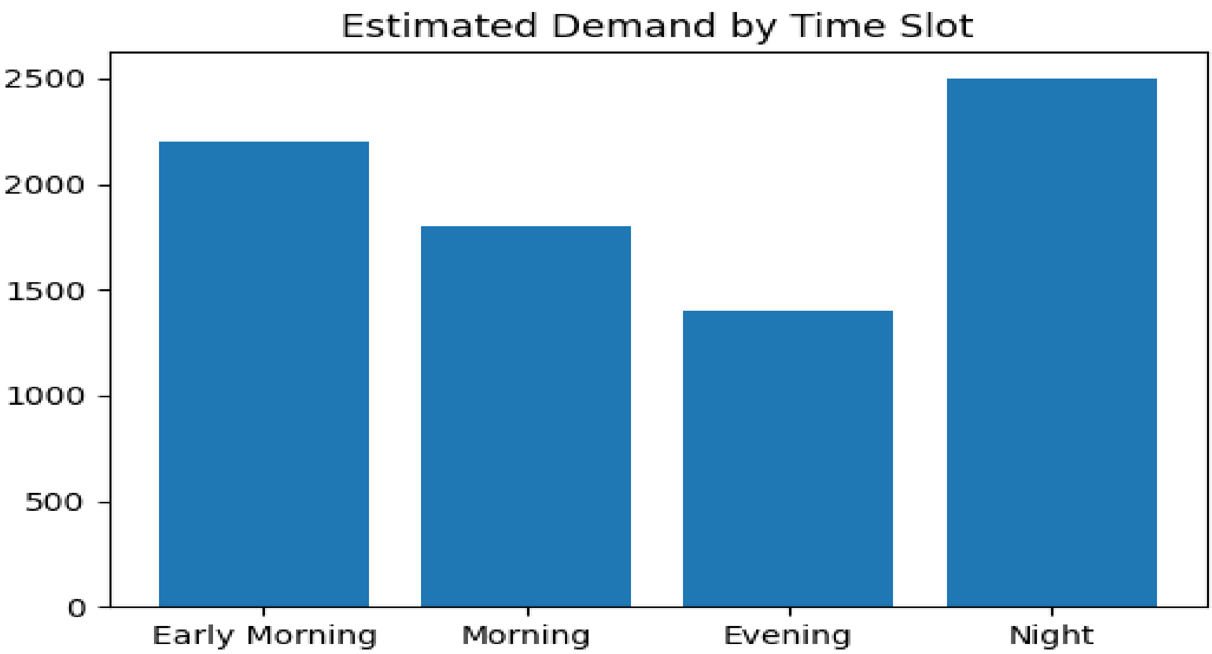
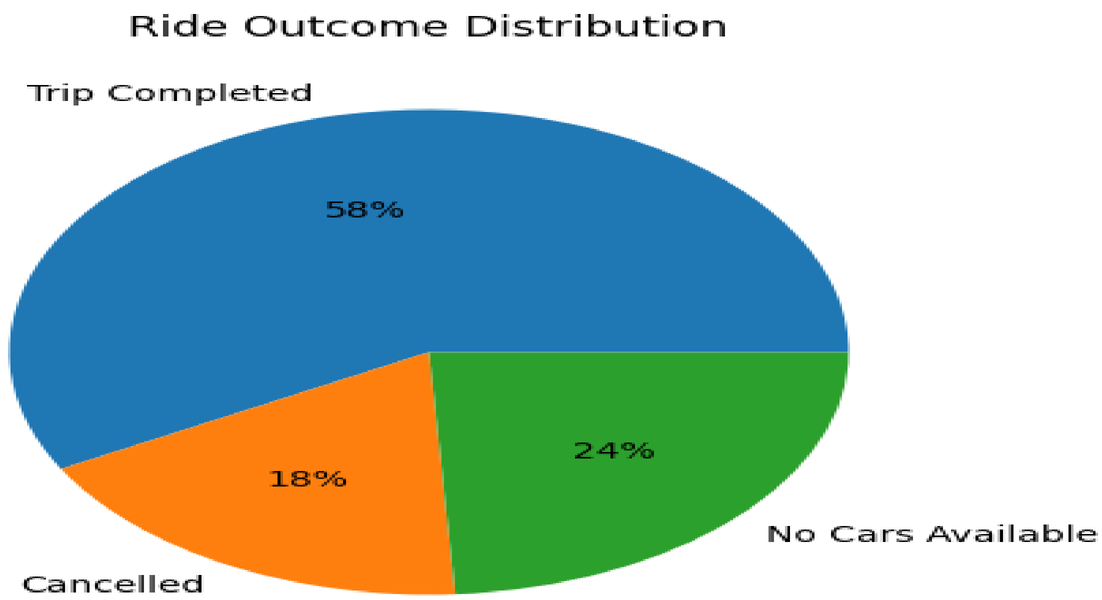
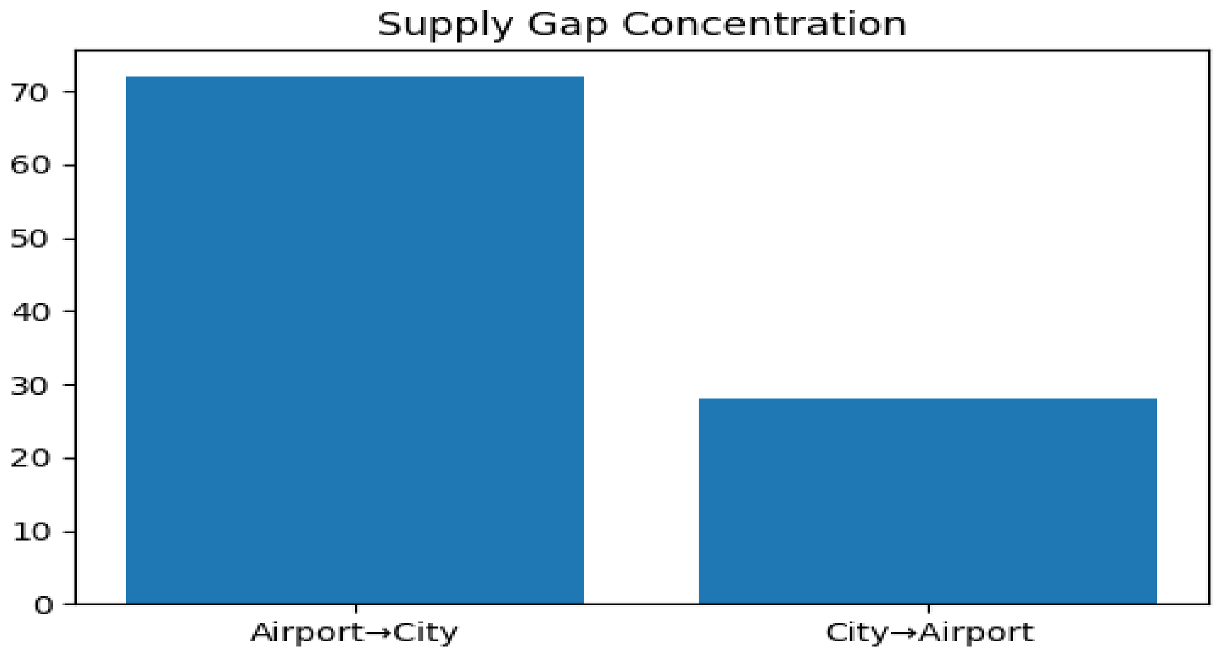
Uber Supply Demand Gap Analysis Insights Report

This report summarizes the exploratory data analysis performed on Uber request data to identify demand–supply gaps, operational bottlenecks, ride completion trends, cancellation behavior, and opportunities for service optimization. Visual insights were used to evaluate request patterns and customer experience outcomes.




Ride Outcome Distribution
Trip Completed
58%
18%
24%
No Cars Available
Cancelled


Supply Gap Concentration
70
60
50
40
30
20
10
0
Airport→City
City→Airport

Professional Conclusion

The analysis identified clear supply–demand imbalances across different request periods and ride outcomes. Peak-hour demand created operational pressure, leading to increased cancellations and unfulfilled requests in specific pickup locations. The distribution of ride statuses indicates opportunities to improve driver allocation and reduce customer wait time. The findings suggest that proactive driver positioning, demand forecasting, and dynamic resource planning can improve ride completion rates and overall customer satisfaction. Monitoring peak periods and optimizing supply allocation may help reduce service gaps and improve operational efficiency. Overall, this project demonstrates how Exploratory Data Analysis (EDA), SQL-based insights, and visualization techniques can convert raw transportation data into meaningful business decisions and measurable operational improvements.

Key Recommendation

Implement demand prediction models, improve driver availability during peak hours, and continuously monitor ride performance using dashboards for better decision-making.